

INDEPENDENT SAMPLES T-TEST

This section explores the independent-samples t-test as a statistical method for comparing means between two groups or treatment conditions. It explains the two main research designs used for this test, namely independent groups and repeated measures, and clarifies how hypotheses are stated and tested. The section also presents the structure of the independent-measures t-statistic, including the roles of the estimated standard error, the pooled variance, and the degrees of freedom. Finally, it discusses key assumptions, confidence intervals, and example applications to support the correct interpretation of results.

Introduction (Gravetter, 2021; Brase et al., 2023)

We have focused on inferential statistics using a single sample to draw conclusions about a single population. However, real-world research often involves comparing multiple samples or treatment conditions. For example:

- A social psychologist might examine differences in political attitudes between Millennials and Generation X adults.
- An educational psychologist could compare two teaching methods for math.
- A clinical psychologist might evaluate therapy effectiveness by comparing patients' depression scores before and after treatment.

These scenarios typically involve comparing the means of two data sets.

Research designs for collecting these data fall into two main categories:

1. **Independent Groups:** Here, data come from two distinct groups of participants. It could involve comparing Millennials to Generation X in terms of political attitudes or comparing academic performance between students with and without laptops. This approach is known as an independent-measures research design or a between-subjects design. It emphasizes comparing two groups using separate, independent samples.
2. **Repeated Measures:** This design collects data from the same participants at different times. For example, a researcher might measure depression in patients before starting therapy and again six weeks later. Or, a psychologist studying moral development might assess children's moral judgments at age 5 and then again at age 10.

The **independent-measures, or between-subjects, design** involves distinct participant groups for each treatment condition or population and is structured to enable comparisons across groups. This method uses two separate samples to represent the two populations or treatment conditions being analyzed.

The Hypotheses for an Independent-Measures Test

An independent-measures study aims to determine whether there's a significant difference in mean scores between two populations or treatment conditions. We use μ_1 to represent the mean of the first population and μ_2 for the second population. The difference between these means is calculated as $\mu_1 - \mu_2$. Consistent with hypothesis testing principles, the **null hypothesis** posits no difference between these means, symbolically written as:

$$H_0 : \mu_1 - \mu_2 = 0$$

It states that there's no difference between the population means. An alternative way to express this null hypothesis is $\mu_1 = \mu_2$ but stating it as a difference equaling zero is more useful for calculating the t-statistic, as it provides a clear numerical target (zero) for testing.

On the other hand, the **alternative hypothesis** suggests that there is a significant difference between the two population means, which can be written as:

$$H_1 : \mu_1 - \mu_2 \neq 0$$

Another way to express the alternative hypothesis is simply $\mu_1 \neq \mu_2$ emphasizing that the two means are not equal.

The Formulas for an Independent-Measures Hypothesis Test

The independent-measures hypothesis test uses a specific version of the t-statistic to compare two samples, reflecting differences between two populations or treatment conditions. This version of the t-statistic, distinct from the one introduced earlier for single-sample scenarios, is aptly named the independent-measures t-statistic to highlight its application in studies involving two independent groups.

The basic structure of the t-statistic is the same for both the independent measures and the single-sample hypothesis tests. In both cases:

$$t = \frac{\text{actual difference between sample data and the hypothesis}}{\text{expected difference between sample data and hypothesis with no treatment effect}}$$

The independent-measures **t** is a two-sample t that doubles all the elements of the single-sample t formulas:

$$t = \frac{\text{sample mean} - \text{population mean}}{\text{estimated standard of error}} = \frac{M - \mu}{S_M}$$

The independent-measures t-statistic evaluates the difference between two population means by using the difference between two sample means:

$$t = \frac{\text{sample mean difference} - \text{population mean difference}}{\text{estimated standard error of sample mean difference}} = \frac{(M_1 - M_2) - (\mu_1 - \mu_2)}{S_{(M_1 - M_2)}}$$

In the independent-measures t-statistic formula, we calculate $M_1 - M_2$ using the mean values from our sample data. This difference in sample means tests the null hypothesis, which typically proposes no difference between the population means $\mu_1 - \mu_2$. Due to this assumption, the formula simplifies since the hypothesized difference is zero, leaving us with the formula:

$$t = \frac{\text{sample mean difference}}{\text{estimated standard error}}$$

This version of the t-statistic is a straightforward ratio: the numerator is the actual observed difference between our sample means, and the denominator is the estimated standard error, which is the difference we'd expect by random chance.

Each t-score formula includes a standard error in the denominator, which estimates the expected error between the sample statistic and the population parameter. In the single-sample t formula, this standard error is for a single-sample mean and is symbolized as S_M . The independent-measures t formula estimates the

expected error for the difference between sample means. $M_1 - M_2$ and the hypothesized difference in population means $\mu_1 - \mu_2$. Here, the standard error of the sample mean difference is denoted as $S_{(M_1 - M_2)}$.

Calculating the Estimated Standard Error

To calculate the estimated standard error for the difference between two sample means $S_{(M_1 - M_2)}$, we must understand that each sample mean M_1 and M_2 is an estimate of its respective population mean (μ_1 and μ_2) and each has its own errors. This error represents the difference between the sample mean and the true population mean, which is unknown.

Each sample mean is accompanied by its estimated standard error, which measures the error.

For the first sample mean M_1 the estimated standard error is $S_M = \sqrt{\frac{s_1^2}{n_1}}$

and for the second sample mean M_2 it is $S_M = \sqrt{\frac{s_2^2}{n_2}}$

In the case of the independent-measures t-statistic, we're interested in the total error when using two sample means to estimate two population means. To get the total error, we calculate the error for each sample separately and then sum them. The formula for the estimated standard error of the difference between the two sample means is therefore:

$$S_{(M_1 - M_2)} = \sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}$$

This equation combines the individual errors from both samples into a single measure of the estimated standard error of the difference between them. This estimated standard error is used in the independent-measures t-statistic to help determine whether the observed difference between sample means is statistically significant.

Pooled Variance

The formula above accurately captures the standard error for an independent-measures t-test, but only when the two samples have the same size. If the sample sizes differ, this formula becomes biased and unsuitable for use. The bias arises because the formula gives equal weight to both sample variances, which isn't appropriate when sample sizes vary. It is because, according to the law of large numbers, larger samples provide more accurate estimates of population parameters than smaller samples.

To correct this bias when sample sizes are unequal, statisticians use a method that combines the two sample variances into one pooled variance. This pooled variance is calculated by averaging the sample variances, giving greater weight to larger samples. This adjustment ensures that the pooled variance is a more precise estimate of the population variance, considering the different sample sizes.

Recall that when there is only one sample, the sample variance is computed as:

$$s^2 = \frac{SS}{df}$$

In the independent-measures t-test, we use two sums of squares (SS), one from each sample, and each has its own degrees of freedom (df). To calculate the pooled variance, we combine the SS values and divide them by the total degrees of freedom across both samples. The formula for pooled variance looks like this:

$$\text{pooled variance} = s_p^2 = \frac{SS_1 + SS_2}{df_1 + df_2}$$

When dealing with just one sample, the variance is the sum of squares (SS) divided by its degrees of freedom (df). But with two samples, we add up the SS from both samples and then divide by the sum of their degrees of freedom to find the pooled variance. This pooling method gives more weight to the larger sample, reflecting its greater impact on the accuracy of the variance estimate. This concept is crucial because the larger the sample, the more reliable the sample variance is as an estimate of the true population variance, according to the law of large numbers.

The Final Formula and Degrees of Freedom

The complete formula for the independent-measures t-statistic is as follows:

$$t = \frac{(M_1 - M_2) - (\mu_1 - \mu_2)}{S_{(M_1 - M_2)}}$$

The degrees of freedom for the independent-measures t-statistic is the sum of the degrees of freedom from both samples. Equivalently, the df value for the independent-measures t-statistic can be expressed as:

$$df = n_1 + n_2 - 2$$

	Sample Data	Hypothesized Population Parameter	Estimated Standard Error	Sample Variance
Single-sample <i>t</i> statistic	M	μ	$\sqrt{\frac{s^2}{n}}$	$s^2 = \frac{SS}{df}$
Independent-measures	$(M_1 - M_2)$	$(\mu_1 - \mu_2)$	$\sqrt{\frac{s_p^2}{n_1} + \frac{s_p^2}{n_2}}$	$s_p^2 = \frac{SS_1 + SS_2}{df_1 + df_2}$

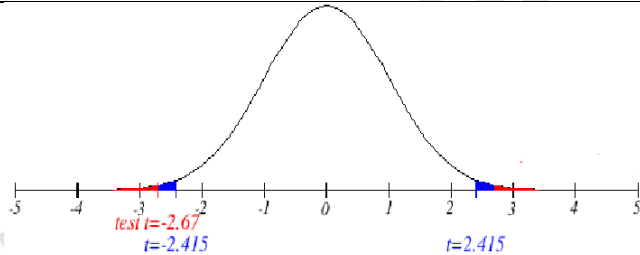
Figure 1. The basic elements of a t-statistic for the single-sample *t* and the independent-measures *t*

Sample Problem: Two-tailed

Research has shown that people are more likely to exhibit dishonest and self-interested behavior in the dark than in a well-lit environment. In one experiment, participants were given a set of 20 puzzles and paid 50 cents for each puzzle solved within a 5-minute period. However, participants reported their performance, and there was no clear way to verify their honesty. Thus, the task provided a clear opportunity to cheat and receive undeserved money. One group of participants was tested in a dimly lit room, and a second group was tested in a well-lit room. The reported number of solved puzzles was recorded for each individual. Test at $\alpha = .05$.

Number of Solved Puzzles			
Well-Lit room		Dimly Lit Room	
11	6	7	9
9	7	13	11

Number of Solved Puzzles			
Well-Lit room		Dimly Lit Room	
4	12	14	15
5	10	16	11
$n_1 = 8$		$n_2 = 8$	
$M_1 = 8$		$M_2 = 12$	
$SS_1 = 60$		$SS_2 = 66$	

Hypotheses	$H_0 : \mu_1 - \mu_2 = 0$ $H_1 : \mu_1 - \mu_2 \neq 0$
Alpha	$\alpha = .05$, two-tailed
df	$df = n_1 + n_2 - 2 = 8 + 8 - 2 = 14$
t-critical	± 2.145
t-statistic	$s_p^2 = \frac{SS_1 + SS_2}{df_1 + df_2} = \frac{60 + 66}{7 + 7} = \frac{126}{14} = 9$ $S_{(M_1 - M_2)} = \sqrt{\frac{s_p^2}{n_1} + \frac{s_p^2}{n_2}} = \sqrt{\frac{9}{8} + \frac{9}{8}} = \sqrt{2.25} = 1.50$ $t = \frac{(M_1 - M_2) - (\mu_1 - \mu_2)}{S_{(M_1 - M_2)}} = \frac{(8 - 12) - 0}{1.5} = \frac{-4}{1.5} = -2.67$
Decision	$t = -2.67$ falls beyond the critical region of ± 2.145 ; therefore, we reject H_0
Graph	
Result	The students who were tested in a dimly lit room reported higher performance scores ($M = 12, SD = 2.93$) than the students who were tested in the well-lit room ($M = 8, SD = 3.07$). The mean difference was significant, $t(14) = -2.67, p < .05$

Sample Problem: One-tailed

When setting up a study comparing two groups, a researcher often has a specific outcome in mind. Take the cheating study as an example: the expectation is that students in a dimly lit room will report higher scores than those in a well-lit room. This kind of expected outcome can lead to a one-tailed test, which is a directional hypothesis test. A one-tailed test can result in the rejection of the null hypothesis with a smaller mean difference than would be required by a two-tailed test, which looks for any significant difference, regardless

of direction. Because of this, one-tailed tests are appropriate when there's a strong theoretical basis or prior research supporting a directional prediction.

Let's revisit the scenario described in the earlier example, where a researcher is investigating how lighting affects dishonest behavior. The hypothesis is that students in a well-lit room are less likely to cheat than those in a dimly lit room. This directional prediction will shape how the hypotheses are stated and how the critical region for the one-tailed test is determined using the independent-measures t-statistic. For this test, we will use $\alpha = .01$.

Hypotheses	$H_0 : \mu_{Well-Lit} \geq \mu_{Dimly Lit}$ $H_1 : \mu_{Well-Lit} < \mu_{Dimly Lit}$
Alpha	$\alpha = .01$, one-tailed
df	$df = n_1 + n_2 - 2 = 8 + 8 - 2 = 14$
t-critical	-2.624
t-statistic	$s_p^2 = \frac{SS_1 + SS_2}{df_1 + df_2} = \frac{60 + 66}{7 + 7} = \frac{126}{14} = 9$ $S_{(M_1 - M_2)} = \sqrt{\frac{s_p^2}{n_1} + \frac{s_p^2}{n_2}} = \sqrt{\frac{9}{8} + \frac{9}{8}} = \sqrt{2.25} = 1.50$ $t = \frac{(M_1 - M_2) - (\mu_1 - \mu_2)}{S_{(M_1 - M_2)}} = \frac{(8 - 12) - 0}{1.5} = \frac{-4}{1.5} = -2.67$
Decision	$t = -2.67$ falls beyond the critical region of -2.624 therefore, we reject H_0
Graph	

References

- Brase, C. H., Brase, C. P., Dolor, J. M., & Seibert, J. A. (2023). *Understandable statistics: Concepts and methods* (13th ed.). Cengage.
- Gravetter, F. J., Wallnau, L. B., Forzano, L. A., & Witnauer, J. E. (2021). *Essentials of Statistics for the Behavioral Sciences* (10th ed.). Cengage.